
Algorithm Processing of SOAP Descriptors with Dimensionality Reduction and Locality Sensitive Hashing

Input:

SOAP matrix $\mathbf{S} \in \mathbb{R}^{N \times d}$ for each configuration in a dataset of X configurations

Target reduced dimension D

Bin width w for hash computation

Number of hash functions H

Random bias vector $b \sim \mathcal{U}(-w, w)$

Output: Buckets containing similar configuration vectors

Step 1: Flattening SOAP Descriptors

for each configuration $x \in X$ **do**

 Compute SOAP matrix $\mathbf{S} \in \mathbb{R}^{N \times d}$

 Flatten S into a configuration vector $c \in \mathbb{R}^{(N \times d)}$

end for

Stack all configuration vectors to form the configuration matrix $C \in \mathbb{R}^{X \times (N \times d)}$

Normalize C row-wise

Step 2: Dimensionality Reduction (PCA)

for each configuration vector $c \in C$ **do**

 Compute reduced vector $c' = PCA(c) \in \mathbb{R}^D$

end for

Stack all reduced vectors to form $C' \in \mathbb{R}^{X \times D}$

Normalize C' row-wise for stable hashing

Step 3: Locality-Sensitive Hashing (LSH) with Max-Value Bucketing

3.1 Generate Hash Functions

Generate a random projection matrix: $W_H \in \mathbb{R}^{D \times N_H}$, $W_H(i, j) \sim \mathcal{N}(0, 1)$

3.2 Compute Hash Values for Each Vector

for each row vector $c'_i \in C'$ **do**

 Project the vector onto the random hyperplanes: $h_i = c'_i W_H^T$

 Apply binning with bias $b \sim \mathcal{U}(-w, w)$: $H_i = \lfloor \frac{h_i + b}{w} \rfloor$

end for

3.3 Assign Buckets Using Max Value

for each row vector $c'_i \in C'$ **do**

 Assign the vector to a bucket using the max hash value: $n_i = \max(H_i)$

 Store c'_i in the bucket labeled by n_i

end for

Step 4: Output Buckets

Store all buckets in a dictionary where keys are hash values, and values are corresponding configuration vectors.

Return Buckets containing similar configuration vectors
